# Shahnewaz Leon

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*Eight years of backend engineering building production microservices, data pipelines, and real-time search systems. PhD at NC State with published work in learned code retrieval, sequential prediction, and LLM evaluation at CHI, HCII, and VL/HCC.*

## EDUCATION

**Ph.D. in Computer Science, North Carolina State University, Raleigh, NC, USA *(CGPA: 4.0/4.0)*** **Expected Dec 2026**
**Dissertation:** *Modeling Code Navigation in Individual, Social, and Collective Context*
**Research Focus:** Learned Ranking, Sequential Prediction, Predictive Behavioral Modeling, LLM Evaluation

**M.Sc. in Computer Science, North Carolina State University, Raleigh, NC, USA *(CGPA: 4.0/4.0)*** **Dec 2024**

**B.Sc. in Information Technology, Jahangirnagar University, Dhaka, Bangladesh *(CGPA: 3.78/4.0)*** **Dec 2013**

## SKILLS

**Programming Languages:** Python, C++, Java, SQL, JavaScript, NodeJS, Linux Shell Scripting
**ML & AI:** PyTorch, TensorFlow, Scikit-Learn, Transformers, Pandas, NumPy
**Scientific Computing:** SciPy, NumPy, Jupyter, Matplotlib, Seaborn
**Distributed Systems & Backend:** Microservices, Event-Driven Architecture, AWS, Docker, Kubernetes, Terraform, Kafka
**Data Stores:** MySQL, Postgres, Elasticsearch, Redis, MongoDB, DynamoDB, Oracle
**Engineering Practice:** CI/CD with GitHub Actions, Jenkins, AWS CodeBuild. PyTest, JUnit, Mocha, Chai. GitHub Copilot.

## RESEARCH EXPERIENCE

**Graduate Research Assistant *(NC State University, USA)*** **Aug 2022 - Present**

- **Graph Search for Code Retrieval with Learned Ranking:** Designed a graph-augmented learned ranking algorithm for code retrieval, combining beam search with structured-perceptron optimization over lexical, structural, and hierarchical features; achieved MRR 0.94 and Recall@1 0.90 on large repositories, against 0.67 / 0.55 for a fine-tuned UniXcoder, evaluated on 541 CodeSearchNet query pairs across 6 Python repositories.
- **Markov Priors for Cold Start Navigation:** Built a sequential prediction architecture using Markov state transitions and call-graph reachability for cold-start recommendation; achieved 0.804 Hit@10; four prior sessions suffice to exceed the within-session baseline. Outperformed Qwen2.5-Coder-32B (0.644) and Gemini 2.5 Pro (0.640) on matched candidate pools with controlled 8,000-token budgets and latency profiling.
- **Collaborative Code Navigation Prediction:** Engineered a sequence model incorporating implicit cues from teammates' recent navigation and explicit cues from team conversation, fused with graph topology; predicted 81.5% of team navigations (up to 57.1% more than individual baseline), reducing unknown predictions from 33.63% to 21.06%, with 9 of 10 teams improving.
- **Wildfire Detection (CSC 722):** Built a transfer learning pipeline for wildfire detection from aerial imagery; evaluated Swin V2, Inception-v3, ResNet-50, and ViT on the FLAME dataset with GAN-based augmentation and ensembling; best result 78.0% accuracy, F1 0.738.

## SELECTED PUBLICATIONS

1. **CHI 2026, Best Paper Honorable Mention.** "Where Will They Click Next?" With S. K. Kuttal.
2. **HCII 2026.** "Learning to Navigate (L2N)." With Z. Zhang, P. Tadepalli, S. K. Kuttal.
3. **HCII 2026.** "What Teams Seek and How They Seek It." With A. R. Desai, S. Kanth, N. Mahmoud, S. K. Kuttal.
4. **VL/HCC 2026.** "Cross-Session Collective Priors." With S. K. Kuttal.

## WORK EXPERIENCE

**Senior Software Engineer *(Monstarlab, Bangladesh)*** **Aug 2017 - Jul 2022**

- **Recruitment Platform:** Built a serverless pipeline on AWS Lambda and EventBridge; automated job posting ingestion via Puppeteer and CasperJS, application processing via a Spring Boot data service, and outreach via Twilio and SendGrid.
- **Genetic Health Platform:** Built backend services for a consumer genetic testing platform; implemented audit trail pattern for sensitive health and disease-risk data; integrated Stripe for subscription billing.
- **APIs and Infrastructure:** Designed and delivered REST APIs in Java Spring Boot and PHP across client products; configured CI/CD with GitHub Actions and AWS CodePipeline; optimized MySQL and PostgreSQL through query tuning and indexing.
- **Technical Leadership:** Led backend teams of three to five engineers; defined HLD and LLD, conducted technical specification reviews, and mentored junior engineers.

**Backend Developer *(Ekhanei.com, Bangladesh)*** **Apr 2016 - Jul 2017**

- Developed REST API microservices in PHP (Slim, Silex, Lumen); deployed Elasticsearch for real-time search indexing; monitored platform health via Kibana; built a content moderation microservice.

**Instructor of Record *(CSC 216, NC State University)*** **May 2026 - Aug 2026**

- Sole instructor for Software Development Fundamentals; delivered in-class lectures; managed teaching assistants and mentored student group projects, graded projects and exams.